CRNet: Channel-Enhanced Remodeling-Based Network for Salient Object Detection in Optical Remote Sensing Images

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***Abstract*— Despite the remarkable progress made by the salient object detection of natural sensing images (NSI-SOD), the complex background and scale diversity issues of remote sensing images (RSIs) still pose a substantial obstacle. In this study, we build an end-to-end channel-enhanced remodeling-based network (CRNet) for optical RSIs (ORSIs) to highlight salient objects through feature augmentation. First, the backbone convolutional block is used to suggest the fundamental characteristics. Then, we use the channel enhance module (CEM) to enhance the shallow features. CEM primarily relies on the channel attention (CA) mechanism and uses a no-downscaling strategy to produce local cross-channel interaction, which lowers model complexity while enhancing extraction performance. Meanwhile, we use the redefined feature module (RFM) to reconstruct the deep features and generate global attention features by dimensional transformation and feature relationship aggregation to achieve the role of locating salient targets. Finally, the cascade combines the multiscale features to provide the final saliency map. To further enhance the representational power of the network, we use a hybrid loss function to improve performance. The proposed approach outperforms current state-of-the-art (SOTA) methods, as shown by several experiments on three available datasets. The source code of the proposed CRNet is available publicly at https://github.com/hilitteq/CRNet.git.**

***Index Terms*— Channel enhance module (CEM), optical remote sensing images (RSIS), redefined feature module (RFM), salient object detection (SOD).**

I. Introduction

ACCORDING to the visual attention (VA) mechanism [1], when faced with a scene, people naturally process the areas of interest and choose to ignore the areas of indifference. These areas of interest to people are known as salient regions. Salient object detection (SOD) is a technique for extracting the salient areas in a picture by imitating the human visual system using intelligent algorithms.

As foundational work in compute vision (CV), SOD [2] is committed to modeling the mechanisms of VA. It gives more valuable information for subsequent in-depth studies, such as image segmentation [3], target tracking [4], image quality assessment [5], image retargeting [6], and image classification [7], [8], [9], by filtering image data and extracting relevant regions. In the field of remote sensing, the SOD of remote sensing images (RSI-SOD) has several potential uses in agriculture, military, and disaster relief, including crop detection [10], building extraction [11], port detection [12], ship detection [13], and airport detection [14].

The distinctiveness of ORSIs [15] results from the fact that they are recorded between 400 and 740 nm by aerial sensors and satellites overhead in red, green, blue (RGB) format. Compared with natural scene images (NSIs), ORSIs present the following difficulties in detection. First, there is the issue of scale diversity. The detection targets of RSI-SOD are aircraft, islands, ships, buildings, and rivers that attract human attention. In addition to their size differences, these targets have different target scale sizes due to the shooting distance. Second, there is the issue of small objects. Small objects account for fewer pixels in the whole image, making it more difficult to learn during the training of the network. Moreover, the network model is oriented to large objects in the process of learning ORSIs, while small objects are easily ignored, leading to poor detection results. Third, there is the issue of background complexity. ORSIs cover a large field of view and allow for a wide variety of complex background information,

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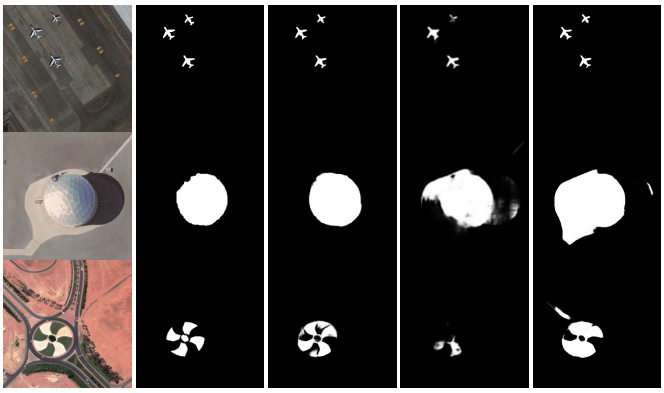
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(a) (b) (c) (d) (e)

Fig. 1. Visual examples of complicated scenarios in ORSIs. (a) ORSIs. (b) Ground truth. (c) Proposed CRNet. (d) LVNet. (e) MCCNet.

which can produce strong interference in detecting salient targets. The issues mentioned above are also a breakthrough for RSI-SOD.

In recent years, researchers have given ORSIs considerable attention, which has resulted in a rapid evolution for RSI-SOD. In the field of traditional methods, Huang et al. [16] proposed a detection model based on contrast-weighted dictionary learning (CDL). In the oil tank detection problem, Liu et al. [17] introduced an unsupervised saliency model based on color Markov chains (US-CMCs). US-CMC counteracted the effects of shadows using colored Markov chains, and the model effectively suppressed salient regions except for tanks. However, since traditional methods rely heavily on specific artificial features, they consume too much labor costs and computing costs, resulting in great reduction of detection efficiency.

With the development of deep learning, CNN models can not only reduce the need for handcrafted features but also reduce the reliance on central biased features. Li et al. [18] constructed an end-to-end depth network called LVNet, which suppressed cluttered backgrounds and highlighted objects by featuring L- and V-shaped modules. Fig. 1 illustrates the results of the proposed channel enhancement reconstruction network (CRNet) and LVNet in the complicated scenarios of ORSIs. From Fig. 1, we observe that LVNet delivers blurred results when detecting small objects and fails to obtain complete salient objects, mainly because LVNet fails to integrate multiscale information, resulting in the incompleteness of features at different scales. Li et al. [19] presented the multicontent complementation network (MCCNet), which combined foreground features, edge features, background features, and global image-level information to highlight prominent locations using their correlation complementarity. As can be seen from Fig. 1, MCCNet can locate prominent objects, but affected by complex backgrounds, it is difficult to distinguish background regions that are too similar in color to salient objects, resulting in false detection of redundant background information. To sum up, current ORSI-SOD methods face several issues. First, there is a problem with robustness. The current SOD methods still have limitations when dealing with complex backgrounds and noise information in ORSIs. Second, there is an issue with large-scale ORSIs processing. The current ORSI-SOD methods are mainly based on two smaller public datasets, and there are still challenges in processing large-scale ORSIs. Finally, there is a real-time problem. Some methods require significant computing resources to achieve better detection effects, resulting in longer processing times that make it difficult to meet the real-time requirements.

Inspired by the above discussion, we propose a new CNN-based RSI-SOD CRNet for the specificity of ORSI images. The backbone network extracts the basic features of ORSI images. According to the feature characteristics of each level, targeted enhancement methods are adopted to improve the representational power of features. Specifically, for shallow features, we create a channel enhancement module (CEM), which combines the channel attention (CA) mechanism and local channel interaction. This module helps establish boundary detail while reducing the distraction of complex backgrounds and shadows. For deep features with richer semantics, we propose a redefined feature module (RFM) to achieve complete detection of salient objects by reconstructing global contextual semantic information. The fusion process of the decoder module infers salient objects in a way of gradually recovering the resolution, combining localization and detail filling to complete the detection task. A hybrid loss function is used to supervise the training process.

Our network structure highlights not only salient regions but also fine-grained object details. The main contributions are summarized as follows.

1) We create CRNet to effectively address the problem of complex backgrounds, object scale fluctuations, and boundary distinction in ORSI images. Hierarchical detection not only obtains multiscale information but also fully uses shallow detailed information and high-level rich semantic information.
2) For shallow high-resolution images, our proposed CEM can accurately distinguish boundary pixels at the high-resolution level. This module makes full use of the CA relation to highlight foreground regions and suppress the interference of complex backgrounds. At the same time, a local cross-channel interaction module is used to enhance the distinction between foreground and background regions.
3) For deeply rich semantic features, we propose RFM to fully exploit the global semantic correlation. The way of reconstructing features is beneficial for improving the perception of salient regions and capturing more discriminative feature regions.
4) To validate the effectiveness of CRNet, we compared it with 17 state-of-the-art (SOTA) methods on three challenging ORSI datasets. Our method achieved excellent results in all the five evaluation metrics.

II. Related Work

SOD has flourished in recent years. In this section, we briefly describe the development of NSI-SOD and review the models proposed for specificity with ORSIs.

A. NSI-SOD

In earlier work, Borji et al. [20] established the first saliency model based on the center-surround difference

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mechanism, which helped pave the way for the advancement of saliency target detection. Following that, methods based on manual a priori criteria, such as foreground consistency, hyperspectral information, superpixel similarity, and histograms, began to be used in saliency target detection. Hou et al. [21] suggested the integration of multilayer features by introducing short connections from deep to shallow layers. Moreover, Li and Yu et al. [22] obtained the feature vector for the saliency of a target pixel by feeding the image blocks around that pixel to the network. However, the applicability and generality of these techniques are usually limited. For example, some methods [23], [24] used multilayer perceptrons to predict map scores. The extracted region features must be extracted continuously, resulting in low computational efficiency.

Inspired by the fact that fully convolutional networks (FCNs) can process images at pixel level, Sun and Wang [25] introduced FCN to construct saliency models and achieved good results. Since then, CNN-based approaches have driven a new wave of development in SOD. Wang et al. [26] proposed a staged approach for branch refinement, where the shallow network was used for spatial details and fine structure control, and the deeper sites were used to generate coarse prediction maps. Similarly, Zeng et al. [27] extracted global semantic information and guided the generation of local high-resolution details that were gradually fused to improve the prediction performance. All these algorithms use a phased methodology, with global detection followed by local identification, which offers a good model structure even though the detection performance on the saliency map is not very good.

To enhance the representation of deep features, Wu et al. [28] suggested discarding shallow large-resolution features and iteratively optimizing high-level features using deep generated attention maps. Zhao and Wu [29] presented a pyramidal attention network to improve lower level structural aspects and higher level contextual features with channeled attention and spatial attention, respectively. To better use the location data in higher level features, Hou et al. [21] extended the holistically nested edge detector (HED) architecture with additional short connections. Notably, most methods only focus on deep and shallow feature enhancement or highlighting of object localization, lacking the systematical integration of them.

B. RSI-SOD

Similar to the traditional NSI-SOD, the traditional RSI-SOD relies on manual features. To create the final salient maps, Zhang and Wu [30] used adaptive assigned feature contributions based on low-rank matrices to recover texture features from binary complex wavelet variations. Huang et al. [16] proposed a method based on CDL that involved discriminative dictionary construction, saliency map generation, and multimap fusion. What is more, Zhang et al. [31] applied color information content analysis to ORSIs to create saliency maps based on different color channel intensities. The traditional RSI-SOD method is limited by the feature extraction performance, and the detection effect is not good.

In addition, the recently released CNN-based strategy advances the entire area much further. To further improve the feature, Li et al. [32] constructed cross-path linkages by densely connecting the same path features. Similarly, Zhang et al. [33] suggested a dense attentional flow module to direct the development of high-level feature attention graphs and global perceptual attention to acquire remote semantic contextual correlations. Furthermore, Liu et al. [34] used graph inference to explore ORSIs, assessing global data and obtaining prominent graphs with both space and channels. Cong et al. [35] captured a high-level semantic method for RSI-SOD through a multiscale attention refinement of the underlying information and relational inference. When using position cue cyclic calibration decoding to produce accurate salient maps, Zhou et al. [36] developed an edge-guided cyclic localization network. Zhou et al. [37] addressed the problem of rough edge details in ORSIs through multiscale feature integration of edge cues. Shen et al. [38] proposed to achieve a balance between computational cost and detection performance through dimensionality reduction and multiscale strategies.

As we know, salient objects of ORSIs usually have complex geometric structures, variable scales, and indeterminate quantities and are often characterized by complex backgrounds, lighting shadows, and imaging blur. The specific methods described above propose meaningful solutions for their target properties. However, in the face of mixed detection of multiple types of objects, we believe that there is still a lot of room for improvement in processing multiscale information, location recognition, and object edges. Our proposed CRNet deeply explores the feature information of different scales and makes full use of the features at each scale. Specifically, CEM and RFM are used to differentiate and aggregate feature information of different scales in the encoder and cascaded in the decoder.

III. Proposed Method

The general structure of the suggested CRNet model for RSI-SOD is shown in Fig. 2. The framework consists of two parts: an encoder based on feature improvement and a decoder for stepwise supervised fusion. The encoder module consists of the backbone convolution block and two enhancement modules: CEM and RFM.

A. Overview

Our proposed CRNet architecture is an encoder–decoder system that adds the CEM and RFM after the backbone encoder to enhance feature extraction. Unlike conventional feature encoders that consist of a direct stack of convolutional blocks, we use CEM to perform enhanced extraction of shallow features. Specifically, we implement attention map extraction using a global average pooling layer (GAP) and grouped convolutions. Then, it is stacked with the original features to obtain the feature map regions annotated with saliency. For deep features, RFM is used to reshape the internal structure of features. This module enhances the correlation of global semantic information and focuses on the location of salient

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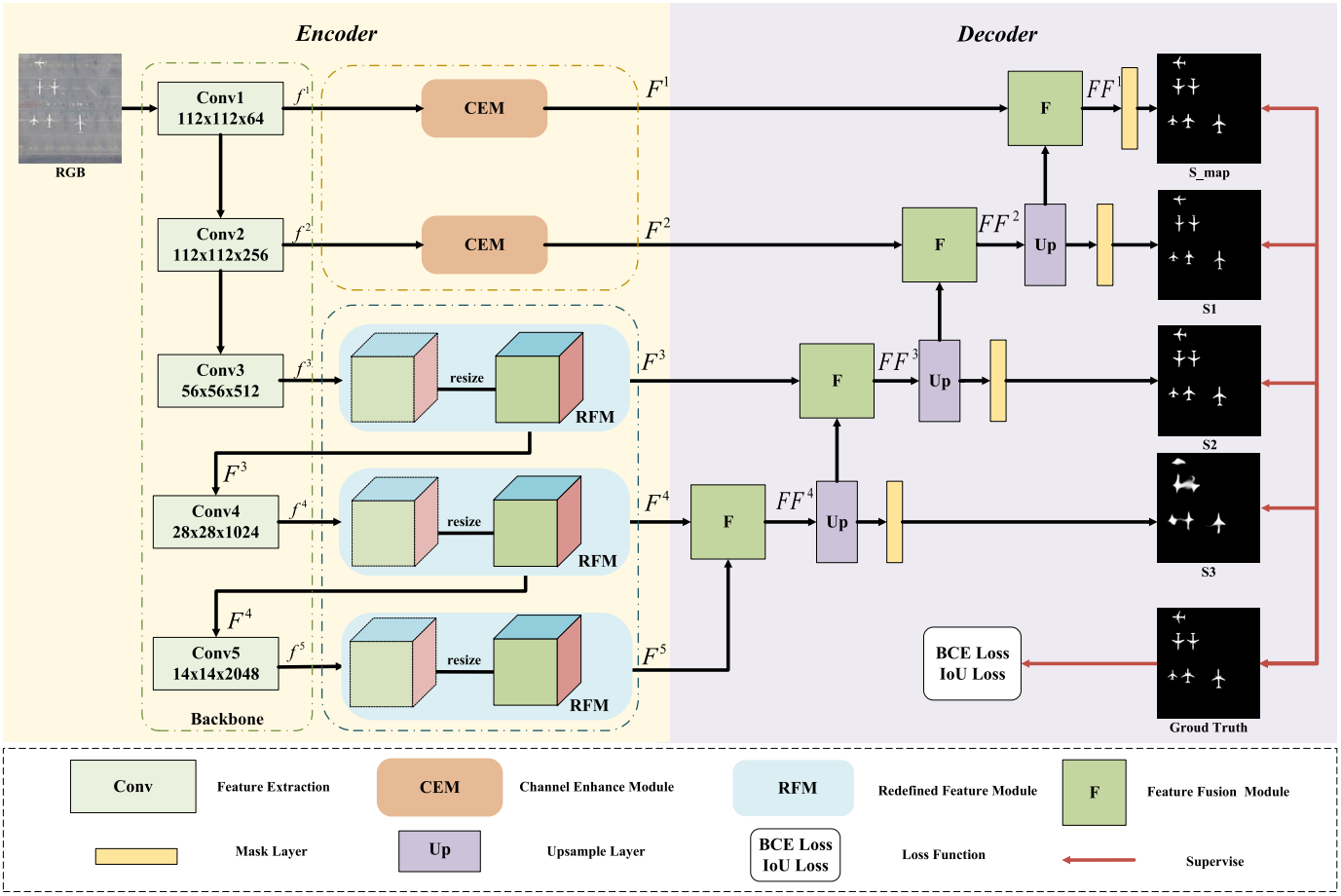


Fig. 2. Overall framework of CRNet includes an encoder based on feature enhancement and a decoder for stepwise supervised fusion. The encoder consists of 1) backbone encoder convolution to extract multilevel features, 2) CEM to obtain efficient CA relations, and 3) RFM to aggregate global relations by feature reconstruction through 2-D transformation. The decoder enhances the feature representation by cascading operation **F** progressively fusing the encoders. Notably, in the training phase, we use a deep supervision strategy with additional map-level and pixel-level monitoring of each side output saliency map. Conv$i$, ($i = 1, 2, \ldots, 5$) distinguish encoder modules of different scales.

objects. In addition, the reshaped features are input into the next convolution block to form nested extraction, which further obtains more favorable feature information while reducing redundancy. Therefore, in the process of feature decoding, the deep and shallow enhanced features are gradually fused, and the final generated salient maps can better locate the salient target locations and refine the target edge details.

B. Feature-Enhanced Encoder

We use Res2Net50 [39] as the fundamental backbone architecture since it can express multiscale characteristics at a finer level of detail without losing generality. In addition, it has strong feature extraction capabilities to handle the scale variation and small target issues of ORSIs. In Fig. 2, we use conv1–conv5 to represent the five convolution blocks of Res2Net50 from shallow to deep, respectively. We define the initial features as $f = \{f^1, f^2, f^3, f^4, f^5\}$, whose numerical superscripts indicate the convolution blocks. We input the input features $I \in R^{C\times H\times W}$ of size $3 \times 224 \times 224$ into the basic backbone features to obtain the corresponding initial features. In general, features with higher resolution and more pixel information are referred to as shallow features, and features with relatively lower resolution and more hidden information are referred to as deep features. First, we feed shallow features f^1 and f^2 into CEM. Then, through the CA mechanism and group convolution, the functions of removing redundant background information and finely expressing foreground details are realized. Finally, the channel enhancement features F^1 and F^2 are generated, and their corresponding scale sizes are $64 \times 112 \times 112$ and $256 \times 112 \times 112$, respectively. For deep features f^3–f^5, RFM is used for feature structure transformation. Reconstruction-enhanced features F^3–F^5 are generated, exploiting the semantic correlation between target pixels and channels. The corresponding scale sizes are $512 \times 56 \times 56$, $1024 \times 28 \times 28$, and $2048 \times 14 \times 14$.

1) CEM: Salient objects are extracted by learning more distinctive characteristics for detection. As previously indicated, attention mechanisms have been extensively exploited in SOD to boost the network's capacity for representation, sharpen focus, and reduce redundant data [40]. As shown in Fig. 2, the shallow features have a higher spatial resolution, which also contains more detailed information and facilitates

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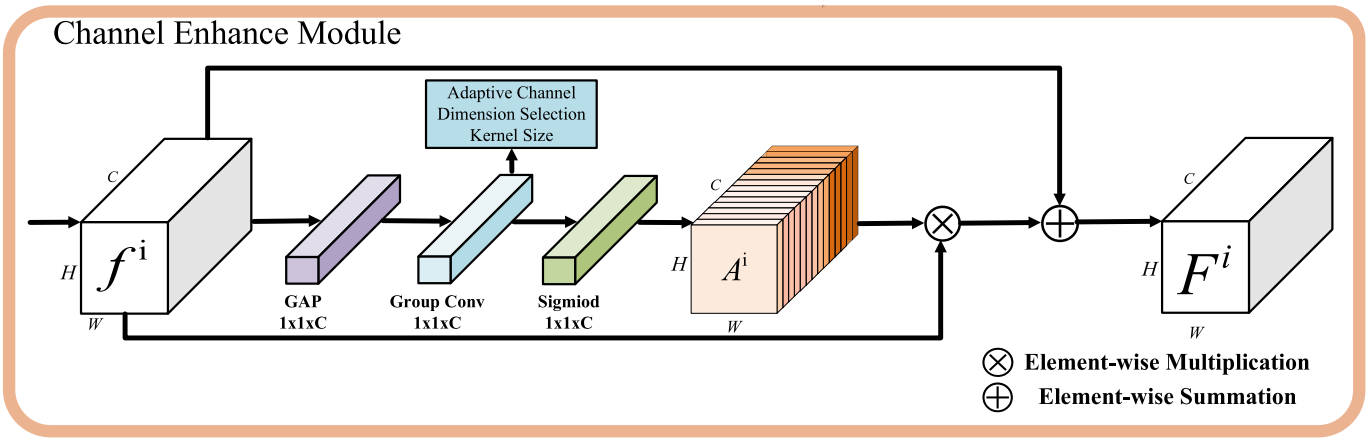


Fig. 3. Proposed CEM module where the group conv adaptively selects the convolution kernel size according to the channel dimension.

the recovery of salient objects at different scales, especially for small objects.

As a result, we apply attention techniques to handle the problem of shifting target scale, direct the recovery of object features, and eliminate the inevitable duplicate information in the backbone encoder. On this basis, we leverage the CEM module to guide the generation of CA maps to enhance shallow features. In the CEM module, we create channel-enhanced feature map F^s by residually concatenating the shallow initial feature f^s and the attention map A^s, where $s \in \{1, 2\}$ indexes the convolution phase (see Fig. 3 for details).

The input features are first subjected to global average pooling, followed by grouped convolution and sigmoid activation to generate the final attention map. Specifically, we perform an adaptive global average pooling operation on f^s, which preserves the spatial information in the convolutional layer by regularizing the channel structure and achieves the acquisition of a single-channel graph containing global information, denoted by $R^s \in R^{C\times1\times1}$

$$R^s = \text{avg_pool}(f^s) \tag{1}$$

where avg_pool$(\cdot)$ denotes adaptive averaging pooling. With the aid of adaptive averaging pooling, dimensionality reduction is avoided by maintaining the same number of channels after completely connected feature information.

We create local cross-channel interaction using the band matrix W_k to train CA using the same learning parameters while taking into account the interaction between various channels and preventing total independence between various channels

$$W_k = \begin{bmatrix} w^{1,1} & \cdots & w^{1,k} & 0 & 0 & \cdots & \cdots & 0 \\ 0 & w^{2,2} & \cdots & w^{2,k+1} & 0 & \cdots & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots w, C-k+1 & \cdots & w^{C,C} \end{bmatrix} \tag{2}$$

where $\omega = \sigma(\text{conv}_{1\times1}(R^s))$, $\sigma(\cdot)$ denotes the sigmoid function, $\text{conv}_{1\times1}$ represents a 1×1 convolution, where k represents the size of the convolution kernel, and C represents the channel dimension. To ensure full coverage of local cross-channel interactions, $\text{conv}_{1\times1}$ uses grouped convolution to ensure that the channel dimension C can be proportional to the convolution kernel size k for a fixed number of groups. A nonlinear function is used with the following equation:

$$k = \frac{\log_2 C + \beta}{\alpha}. \tag{3}$$

In this article α is set to 2 and β is set to 1, and the size of the convolution kernel used for $\text{conv}_{1\times1}$ is determined by the channel dimension C. An expand$(\cdot)$ function is used to expand the learned CA $W_k \in R^{C\times1\times1}$ to generate the CA map $A^s \in R^{C\times H\times W}$, which is calculated as follows:

$$A^s = \text{expand}(W_k). \tag{4}$$

Finally, the obtained CA maps are combined with the initial features to generate the final channel enhancement features F^s, which are calculated as follows:

$$F^s = f^s \otimes A^s \oplus f^s \tag{5}$$

where $\otimes$ denotes elementwise multiplication, and $\oplus$ denotes elementwise summation.

2) RFM: Due to the diverse target scales, different numbers of targets, and complex backgrounds in ORSIs leading to severe redundancy, it is difficult to detect salient objects with accurate locations and detailed edges. As seen in Fig. 1, there are several targets of the same item that have the same visual and semantic characteristics. After identifying one kind of target, similar semantic correlations can help detect targets of the same type. Constructing correlations between different spatial regions helps detect structurally complete targets, and graph-based models are effective in inferring contextual semantic correlations [34]. Therefore, we can enhance the semantic correlations between pixels, locate target locations, and aggregate target structures through graph transformations.

We construct two reshaping operations of the input initial features to ensure the establishment of different spatial relations. The first reshaping is performed on the initial features $f^s \in R^{C\times H\times W}$, where $s \in \{3, 4, 5\}$. First, the 2-D transformation fusion of f^s is performed using channel attention module (CAM) and conv layers, respectively. Second, the

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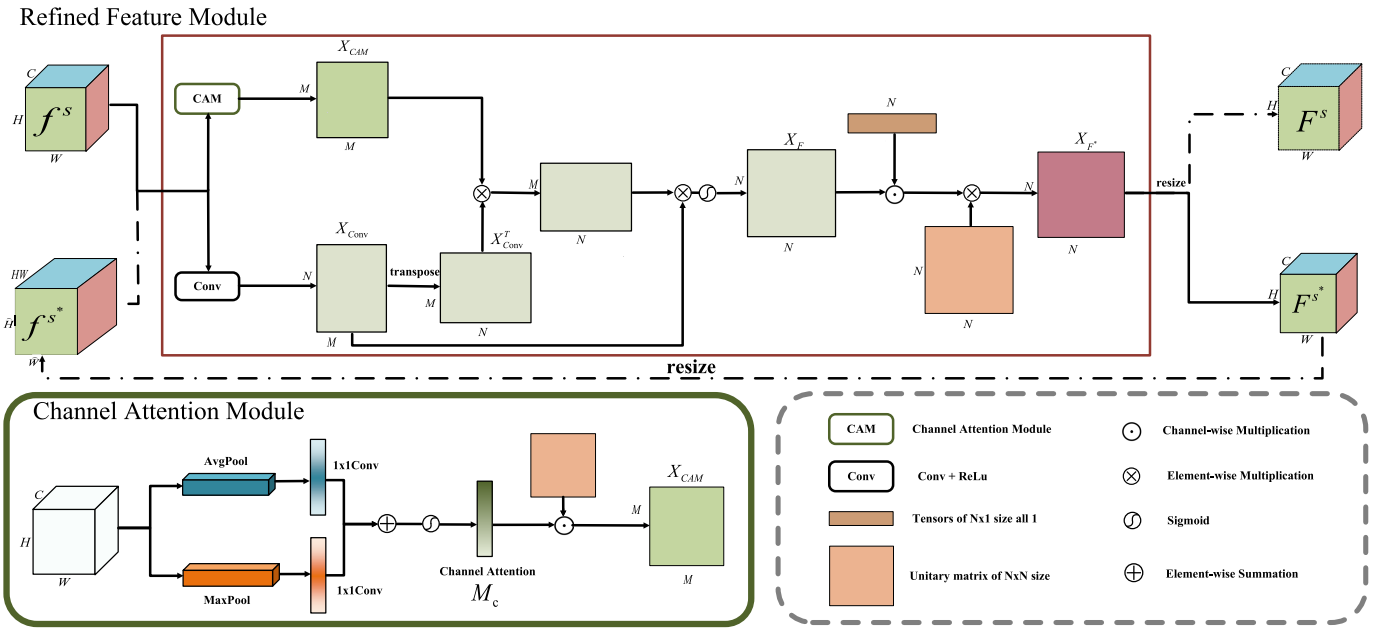


Fig. 4. Proposed RFM module. The solid line indicates the first reconfiguration process and the dashed line indicates the second reconfiguration process. The specific reconfiguration process is shown in the orange box, and the interaction is fused by a 2-D relationship diagram, where the CAM process is shown in the CA module.

fusion features are position-highlighted using the I-unit matrix and the diagonal matrix. Finally, the enhanced 2-D matrix is transformed into 3-D features $F^{s^*} \in R^{C\times H\times W}$. The second reshaping of $F^{s^*} \in R^{C'\times H'\times W'}$ is performed first, and then the above operation is performed to obtain the final secondary enhanced feature $F^s \in R^{C\times H\times W}$.

a) Two-dimensional transformation: The distribution of targets in ORSIs is usually uneven, for instance, objects of rivers and bridges may traverse the entire image. There are also targets with long-distance coverage that spans significant distances at both the ends. No matter how far the object range extends, the objects' integrity means that their semantic correlations are interrelated. Therefore, we can capture the connections between target pixels by establishing semantic correlations between arbitrary spaces. We used CAM and Conv layers for the 2-D transformation of 3-D space, respectively, since 2-D maps are easier to calculate the semantic information between pixels than images with 3-D space.

CAM is the key component of RFM, the details of which are shown in Fig. 4. First, average pooling and maximum pooling are used to compress the global distribution data into pixels to retrieve the fundamental features of an image. Then, using 1×1 convolutional layers, feature smoothing is done independently. Finally, the two features are superimposed to obtain the CA map M_c. Its computational equation is expressed as follows:

$$M_c = \sigma(\text{concat}(\text{conv}_{1\times1}(\text{avg_pool}(f^s), \text{conv}_{1\times1}(\text{max_pool}(f^s))) \quad (6)$$

where f^s represents the initial features corresponding to level s, $s \in \{3, 4, 5\}$. concat$(\cdot)$ represents the feature connectivity along the channel dimension. $\text{conv}_{1\times1}$ is a customized layer that combines a 1×1 convolutional layer and a rectified linear unit (ReLu) nonlinear activation function. avg_pool$(\cdot)$ and max_pool$(\cdot)$ represent the average pooling and the maximum pooling, respectively.

Combining M_c with the I-unit matrix to concentrate on the inner product can help learn a better distance measure for the 2-D graph X_{CAM}. The calculations are specified as follows:

$$X_{\text{CAM}} = M_c \odot I \quad (7)$$

where $\odot$ represents the channelwise multiplication.

b) Enhanced remodeling: In addition to the 2-D map X_{CAM} that contains channel information, we can also get the 2-D map X_{conv} that contains spatial information using a straightforward convolution layer

$$X_{\text{conv}} = \text{conv}_{1\times1}(f^s). \quad (8)$$

After determining the corresponding 2-D graphs X_{CAM} and X_{conv}, we use the Euclidean distance to calculate the similarity graph X_F between the measured two 2-D graphs, the details of which are illustrated in Fig. 4, and the calculation formula is as follows:

$$X_F = \sigma(X_{\text{conv}} \odot X_{\text{CAM}} \odot X_{\text{convT}}) \quad (9)$$

where X_{convT} denotes the result after X_{conv} transposition.

To make the semantic correlation distinction between X_F pixels more obvious and further highlight the saliency goal, the enhanced similarity graph X_{F^*} is formulated as follows:

$$X_{F^*} = X_F \otimes \text{One} \odot I \quad (10)$$

where One represents the full 1 vector that can be multiplied with X_F.

Finally, we use spatial transformation to return the 2-D map $X_{F^*} \in R^{N\times N}$ to its initial 3-D size $F^{s^*} \in R^{C\times H\times W}$. We reconstruct F^{s^*} after resizing $f^{s^*} \in R^{C'\times H'\times W'}$, where

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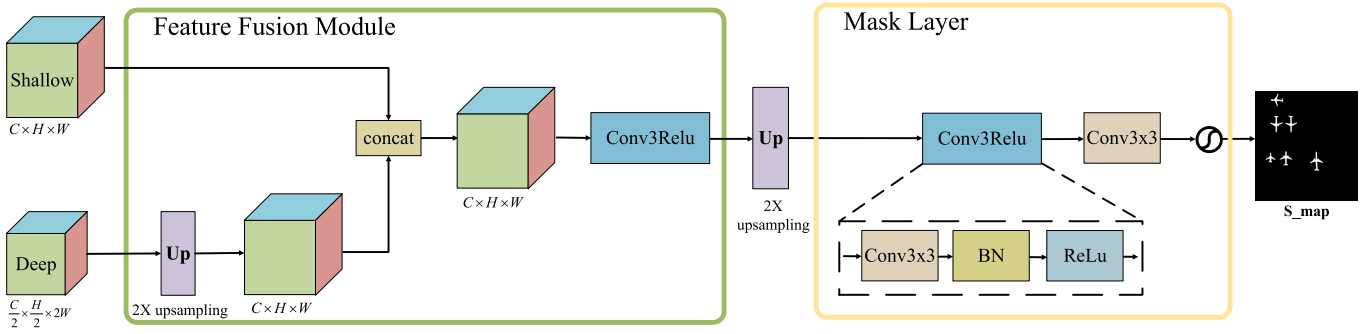


Fig. 5. Process of fusing and binarizing two different scale features in the proposed decoder module to generate a saliency map.

$C \times H \times W = C' \times H' \times W'$, to prevent information loss and maintain the initial total eigenvalues. This reduces the interference of spatial information. We reinput f^{s^*} to the reshaping procedure to finally acquire the secondary reshaping enhancement feature $F^s \in R^{C\times H\times W}$, as illustrated in the dashed flow in Fig. 4.

C. Progressive Fusion Decoder

As shown in Fig. 2, the decoder module performs bottom-up stepwise fusion of the improved features obtained by the encoder to obtain the final saliency map and uses upsampling to unify the spatial resolution of the side outputs. The specific process is shown in Fig. 5. First, we upsample the deep features input to the top-up feature fusion (F) module to ensure uniform size. As shown in Fig. 2, the features generated by the F module are defined as $\{\mathrm{FF}^4, \mathrm{FF}^3, \mathrm{FF}^2, \mathrm{FF}^1\}$. Then, the deep and shallow features are concatenated and input to the Conv3Relu module to obtain the decoded features, including conv3 × 3 (3 × 3 convolution), batch normalization (BN), and ReLu nonlinear activation function. Finally, we upsample the decoded features to unify the size of the side outputs, which are then input to the mask prediction layer to map the saliency score range to [0, 1] using Conv3Relu, conv3 × 3, and sigmoid functions. The prediction layer converts 3-D features into 2-D saliency maps and saliency maps of related layers $\{S1, S2, S3\}$, and outputs the final predicted saliency map S_map on the side.

D. Hybrid Loss Function

The loss function can improve the model performance without increasing the model parameters. In the training phase, we use deep supervision to detect the saliency map of the lateral outputs for fast convergence. As shown in Fig. 2, we fully supervise the saliency map of the lateral output with the red line bidirectional arrows. We integrate pixel-level binary cross-entropy loss (BCE) [41] and map-level intersection over union (IoU) loss [42] to accomplish comprehensive and complementary content augmentation, drawing on the successful use of blending and complementary losses in SOD [43]. The total loss function $\mathcal{L}$ is formulated as

$$\mathcal{L} = L_{\mathrm{bce}}(\mathrm{up}(S), \mathrm{GT}) + L_{\mathrm{iou}}(\mathrm{up}(S), \mathrm{GT}) \tag{11}$$

where $L_{\mathrm{bce}}(\cdot,\cdot)$ represents the BCE loss, $L_{\mathrm{iou}}(\cdot,\cdot)$ represents the IoU loss, S denotes the predicted saliency map, and GT denotes the ground truth. The total loss function is applied to all the lateral outputs so that the deeply supervised strategy with mixed losses not only guides the training process but also improves the final detection accuracy. The overall process of the proposed CRNet method is shown in Algorithm 1.

Algorithm 1 CRNet Model for RSI-SOD

Input: Input an ORSI data $I \in R^{C\times H\times W}$ and ground truth $GT \in R^{C\times H\times W}$.

Output: Predicted salient maps of the test dataset.

1: Set batch size = 6, optimizer Adam (initial learning rate = $5e-5$, minimum learning rate = $5e-6$), epochs ϵ = 50.
2: Obtain the I_{aug} after augment transform. (To simplify the expression, I is used in the main text.)
3: Generate training loader and test loader.
4: **for** $i = 1$ to ϵ **do**
5: Perform Res2Net50 module and generate initial features f^1–f^5.
6: Input f^1 and f^2 to CEM module and generate shallow enhancement features F^1 and F^2. (See 5.)
7: Input f^3–f^5 to RFM module.
8: Perform CAM layer and Conv layer.
9: Flatten each 3-D feature map into 2-D feature vector.
10: Combine diagonal and I-unit matrix to generate deep enhancement features F^3–F^5.
11: Sequentially fuse the enhanced features to produce the final salient map and the side output for supervisory guidance.
12: **end for**
13: Use test dataset with the trained model to get predicted salient maps.

IV. Experiments

A. Experimental Protocol

1) Evaluation Metrics: The performance of our method and the SOTA methods is assessed quantitatively using five widely used metrics [44], including the structural metric (S_α) [45], the (maximum, average, and adaptive) F-measure (F_β, $\beta^2 = 0.3$) [46], the (maximum, average and adaptive) augmented localization metric (E_ξ) [47], the mean absolute

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TABLE I

QUANTITATIVE COMPARISON OF OUR METHOD WITH 16 OTHER SOTA METHODS, INCLUDING FIVE TRADITIONAL NSI-SOD METHODS, THREE CNN-BASED NSI-SOD METHODS, THREE TRADITIONAL RSI-SOD METHODS, AND FIVE CNN-BASED RSI-SOD METHODS, ON TWO PUBLICLY POPULAR DATASETS WITH S_MEASURE, MAXIMUM, MEAN AND ADAPTIVE F-MEASURE, MAXIMUM, MEAN AND ADAPTIVE E_MEASURE, AND MAE. ↑ AND ↓ DENOTE LARGER AND SMALLER, RESPECTIVELY. THE TOP THREE RESULTS ARE MARKED IN RED, BLUE, AND GREEN, RESPECTIVELY

	Methods	Speed	#Param	FLOPs	EORSSD								ORSSD							
		$(fps)\uparrow$	(M)↓	(G)↓	$S\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	Mae ↓	$S\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	Mae ↓
T.N.	$RRWR_{15}$ [50]	0.3	-	-	0.5997	0.4019	0.3707	0.3346	0.6910	0.5959	0.5642	0.1677	0.6835	0.5590	0.5125	0.4874	0.7649	0.7017	0.6949	0.1324
	$HDCT_{16}$ [51]	7	-	-	0.5975	0.5427	0.4034	0.2662	0.7876	0.6388	0.5196	0.1087	0.6197	0.5257	0.4235	0.3722	0.7719	0.6495	0.6291	0.1309
	DSG_{17} [52]	0.6	-	-	0.6428	0.5259	0.4623	0.4019	0.7276	0.6613	0.6196	0.1246	0.7195	0.6238	0.5747	0.5657	0.7912	0.7337	0.7532	0.1041
	SMD_{17} [53]	-	-	-	0.7112	0.5951	0.5502	0.4093	0.7720	0.7309	0.6427	0.0770	0.7640	0.6692	0.6214	0.5568	0.8230	0.7745	0.7682	0.0715
	$RCRR_{18}$ [54]	0.3	-	-	0.6013	0.6013	0.3709	0.3349	0.6898	0.5962	0.5639	0.1644	0.6849	0.5591	0.5126	0.4876	0.7651	0.7021	0.6950	0.1277
C.N.	$EGNet_{19}$ [55]	9	108.07	291.9	0.7444	0.5533	0.5228	0.4526	0.8025	0.7527	0.7118	0.0344	0.7410	0.5816	0.5583	0.5381	0.8008	0.7577	0.7687	0.0630
	$PoolNet_{19}$ [56]	25	53.63	123.4	0.8217	0.7575	0.6432	0.4627	0.9318	0.8215	0.6851	0.0210	0.8403	0.7706	0.6999	0.6166	0.9343	0.8650	0.8124	0.0358
	$MINet_{20}$ [57]	12	47.56	146.3	0.9040	0.8344	0.8174	0.7705	0.9442	0.9346	0.9243	0.0093	0.9040	0.8761	0.8574	0.8251	0.9545	0.9454	0.9423	0.0144
T.R.	VOS_{18} [58]	-	-	-	0.5083	0.2776	0.2115	0.1843	0.5999	0.4895	0.4776	0.2095	0.5366	0.3417	0.2717	0.2633	0.6514	0.5352	0.5826	0.2151
	CMC_{19} [17]	-	-	-	0.5800	0.3272	0.2696	0.2009	0.6818	0.5900	0.4890	0.1057	0.6033	0.3913	0.3454	0.3108	0.7064	0.6417	0.5996	0.1267
	$SMFF_{19}$ [30]	-	-	-	0.5405	0.5208	0.3011	0.2089	0.7768	0.5210	0.5019	0.1434	0.5312	0.4417	0.2684	0.2496	0.7402	0.4920	0.5676	0.1854
C.R.	$LVNet_{19}$ [18]	1.4	-	-	0.8644	0.7824	0.7356	0.6306	0.9276	0.8826	0.8465	0.0145	0.8815	0.8263	0.7995	0.7506	0.9456	0.9259	0.9195	0.0207
	$DAFNet_{21}$ [33]	26	29.35	68.5	0.9166	0.8614	0.7845	0.6427	0.9861	0.9291	0.8446	0.0060	0.9191	0.8928	0.8511	0.7876	0.9771	0.9539	0.9360	0.0113
	$RRNet_{21}$ [35]	16	67.57	255.5	0.9266	0.8785	0.8394	0.7416	0.9684	0.9442	0.9150	0.0074	0.9328	0.8964	0.8720	0.8249	0.9679	0.9547	0.9498	0.0112
	$MCCNet_{21}$ [19]	95	67.65	112.8	0.9327	0.8904	0.8604	0.8137	0.9755	0.9685	0.9538	0.0066	0.9437	0.9155	0.9054	0.8957	0.9800	0.9758	0.9735	0.0087
	$ERPNet_{22}$ [36]	20	77.19	263	0.9252	0.8743	0.8269	0.7170	0.9665	0.9366	0.8994	0.0082	0.9352	0.9036	0.8798	0.8392	0.9738	0.9304	0.9521	0.0114
	$EMFINet_{22}$ [37]	3.7	95.09	353	0.9290	0.8720	0.8486	0.7984	0.9711	0.9604	0.9501	0.0084	0.9366	0.9002	0.8856	0.8617	0.9737	0.9671	0.9663	0.0109
	Ours	17	67.6	255	0.9370	0.8873	0.8653	0.8140	0.9743	0.9618	0.9563	0.0063	0.9389	0.9107	0.8951	0.8695	0.9793	0.9687	0.9711	0.0091

T.N.: Traditional NSI-SOD method; C.N.: CNN-based NSI-SOD method; T.R.: Traditional RSI-SOD method; C.R.: CNN-based RSI-SOD method. The subscript of the method abbreviation corresponds to the year of publication.

error (Mae) [48], the precision–recall curve (PR) [49], and the F-measure curves. We use the assessment tool to help with the assessment.[1]

2) Datasets: On three RSI-SOD datasets that are available to the general public, we train and assess our methodology. The ORSSD dataset [18] consists of 800 challenging ORSIs and their corresponding ground-truth maps, 600 of which are used for training and 200 for testing. The EORSSD dataset [33] is an expansion of the ORSSD dataset that includes more difficult and semantically relevant images. It has a total of 2000 ORSIs, of which 1400 are used for training and 600 for testing. The ORSI-4199 dataset [59] is a challenging dataset with various sizes and scales of objects and background interference. It has a total of 4199 ORSIs, of which 2000 are used for training and 2199 for testing. We use a data enhancement strategy that enriches the training samples using random rotations of 90°, 180°, or 270° to avoid overfitting.

3) Implementation Details: All the experiments are conducted on the PyTorch platform and deployed on workstations equipped with NVIDIA GeForce RTX 3070 GPUs. Our suggested network enables end-to-end training with a uniform input sample size of 224 × 224 for computational effectiveness. Res2Net50 is used to initialize the encoder features while Xavier initializes the remaining features during the training phase. With a batch size of 6, the ADAM algorithm [60] is used to optimize the network. The initial learning rate is $5e-5$, while the minimum learning rate is $5e-6$.

B. Comparison With SOTA

We compare the proposed CRNet to 16 cutting-edge techniques, including eight conventional techniques (RRWR [50], HDCT [51], DSG [52], SMD [53], RCRR [54], VOS [58], CMC [17], and SMFF [30]) and eight CNN-based techniques (EGNet [55], PoolNet [56], MINet [57], LVNet [18], DAFNet [33], RRNet [35], MCCNet [19], ERPNet [36], and EMFINet [37]), to validate its effectiveness. Saliency maps are built using open source code. In the case of not finding the source code and saliency map, we directly use the saliency map provided by MCCNet as a comparison. The NSI-SOD techniques are retrained using the same training set as in this research while keeping the default parameters for all comparisons. At the same time, we are retrained the RSI-SOD comparison method under the same environment configuration.

1) Quantitative Comparison: In Table I, we show the quantitative performance of our method and other SOTA methods on EORSSD and ORSSD. As can be seen from the highlighted values, our method has a clear advantage on the EORSSD dataset, ranking first in the four evaluation metrics, and the other metrics can also achieve second results. In addition to being inferior to MCCNet on the ORSSD dataset, it ranks second in all the evaluation indicators. In addition, we discover that methods created for ORSIs perform noticeably better than those created for NSIs, proving the necessity of designing networks for ORSIs.

Comparison of Table I indicates that the CNN-based technique has a stronger quantitative performance, so we provide the PR and F-measure curves of the CNN-based method on the two datasets, as shown in Fig. 6. Our method surpasses other SOD models under the PR curve and F-measure curve metrics because it is noted that the PR curve of our model is closest to the coordinates (1, 1) and that the region below the F-measure curve of our model is likewise the largest.

According to the results of the ORSSD and EORSSD datasets, we compared our method with the CNN-based competitive SOTA methods on the ORSI-4199 dataset. The comparison results are shown in Table II. From Table II,

[1]https://github.com/MathLee/MatlabEvaluationTools/tree/main/Code

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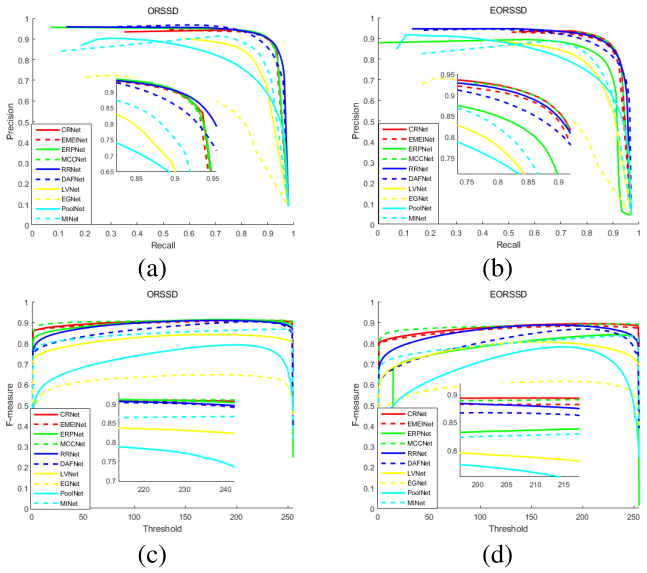


Fig. 6. PR curves and F-measure curves of different methods on the ORSSD dataset and EORSSD dataset, respectively. (a) PR curves of different methods on the ORSSD dataset. (b) PR curves of different methods on the EORSSD dataset. (c) F-measure curves of different methods on the ORSSD dataset. (d) F-measure curves of different methods on the EORSSD dataset.

TABLE II

COMPARISON OF SOTA ON THE ORSI-4199 DATASET, DATA COMPARISON AMONG EIGHT EVALUATION METHODS, THE BEST RESULTS ARE INDICATED IN **Bold**

Methods	$S\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	Mae $\downarrow$
$DAFNet_{21}$ [33]	0.8434	0.8308	0.8648	0.0442
$RRNet_{21}$ [35]	0.7372	0.7349	0.7824	0.0945
$MCCNet_{21}$ [19]	**0.8730**	0.8661	0.9166	**0.0324**
$ERPNet_{22}$ [36]	0.8117	0.8221	0.8867	0.0533
$EMFINet_{22}$ [37]	0.8463	0.8325	0.8945	0.0373
Ours	0.8669	**0.8669**	**0.9176**	0.0335

TABLE III

COMPARISON OF RESULTS ON THE CHALLENGING SCENES BCO, CSO, AND ISO IN THE CHALLENGING ORSI-4199 DATASET. THE BEST RESULTS ARE INDICATED IN **Bold**

Methods	BSO				CSO				ISO			
	$S\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	Mae $\downarrow$	$S\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	Mae $\downarrow$	$S\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	Mae $\downarrow$
$DAFNet_{21}$ [33]	0.8595	0.8939	0.9034	0.0873	0.8404	0.8612	0.8936	0.0926	0.8130	0.8583	0.8740	0.1136
$RRNet_{21}$ [35]	0.6583	0.6847	0.6701	0.2223	0.6450	0.6620	0.6652	0.2198	0.6677	0.6852	0.6741	0.1943
$MCCNet_{21}$ [19]	0.8784	0.9140	0.9169	**0.0644**	0.8784	0.8820	0.9084	**0.0719**	0.8517	0.8931	0.8983	**0.0777**
$ERPNet_{22}$ [36]	0.8452	0.8874	0.8864	0.0947	0.8166	0.8482	0.8720	0.1033	0.7925	0.8506	0.8458	0.1207
$EMFINet_{22}$ [37]	0.8754	0.9147	0.9157	0.0655	0.8549	0.8844	0.9101	0.0704	0.8538	0.8995	0.9009	0.0769
Ours	**0.8802**	**0.9241**	**0.9255**	0.0661	**0.8581**	**0.8906**	**0.9156**	0.0720	**0.8541**	**0.9064**	**0.9100**	0.0802

our method's competitive performance is achieved compared with other methods on the ORSI-4199 dataset. As shown in Table III, for the challenge set in the ORSI-4199 dataset, our method can achieve better detection results in the face of big salient object (BSO), complex salient object (CSO), and incomplete salient object (ISO). Large targets in BSO account for more than half of the image. When the target volume is too large, the integrity of target detection is difficult to maintain. Our method can fully aggregate global semantic context information without destroying object structure while preserving object integrity. The target of CSO is complex, and due to its target texture, structure, and complex boundary, our method is able to achieve fine-grained discrimination

TABLE IV

ABLATION ANALYSIS ON THE EORSSD DATASET

Baseline	CA	CEM	SA	RFM	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$Mae\downarrow$	$S_\alpha\uparrow$
✓					0.8596	0.9546	0.0100	0.9197
✓	✓				0.8580	0.9528	0.0096	0.9194
✓		✓			0.8623	0.9567	0.0096	0.9224
✓			✓		0.8796	0.9728	0.0092	0.9318
✓				✓	0.8790	0.9735	0.0071	0.9276
✓	✓		✓		0.8798	0.9722	0.0078	0.9331
✓	✓			✓	0.8847	**0.9756**	0.0068	0.9364
✓		✓	✓		0.8720	0.9637	0.0076	0.9291
✓		✓		✓	**0.8873**	0.9743	**0.0063**	**0.9370**

TABLE V

CROSS-CHANNEL NUMBER ABLATION IN CEM ON THE EORSSD DATASET

Modules	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$Mae\downarrow$	$S_\alpha\uparrow$
adaptive cross-channel number	**0.8873**	**0.9743**	**0.0063**	**0.9370**
fixed cross-channel number(k=3)	0.8694	0.9656	0.0074	0.9261

in the face of changeable boundaries. The ISO target is incomplete, which increases the difficulty of detection. The method we propose starts with the semantic correlation of the target, respectively, enhances the structural semantics of the background information and the foreground information, and significantly improves the discrimination of the foreground target to achieve the detection task.

2) Qualitative Comparison: Fig. 7 depicts the various visual comparisons between the proposed method and the deep-learning-based methods in a variety of demanding ORSI settings, such as vehicles, rivers, lakes, aircraft, battleships, and ships. The first column is the acquired ORSIs' image, the second column is the corresponding ground-truth maps, and our results are in the third column, followed by the prediction results of the other compared methods. By comparison, our method has the following advantages.

a) Advantages in scenarios with complex backgrounds: The NSI detection algorithms EGNet, PoolNet, and MINet all have inaccurately detected background regions in a row (a) of Fig. 7. Our model, in contrast, can precisely pinpoint the locations of all the targets within a complicated background. ERPNet, MCCNet, RRNet, and LVNet are better at suppressing background redundancy compared with models made for ORSIs. However, they have issues with partial detection errors. Although DAFNet's detection completeness is great, our results show that it lacks clarity on the borders of the details. As can be observed, our method not only effectively reduces background noise in complicated scenes but also accurately detects all the salient targets.

b) Advantages in a target with variable boundaries: Targets with varying borders and variable edge details are detected in (b)–(d) rows of Fig. 7. Except for EGNet which cannot detect all the targets, all the methods can detect two salient objects, but the detected target boundaries are blurred compared with the proposed CRNet. The row (c) comparison demonstrates that LVNet and MINet are unable to identify salient targets, and the detection impact of other comparison methods is not very clear, in contrast to the proposed CRNet, which also has distinct borders that overlap out.

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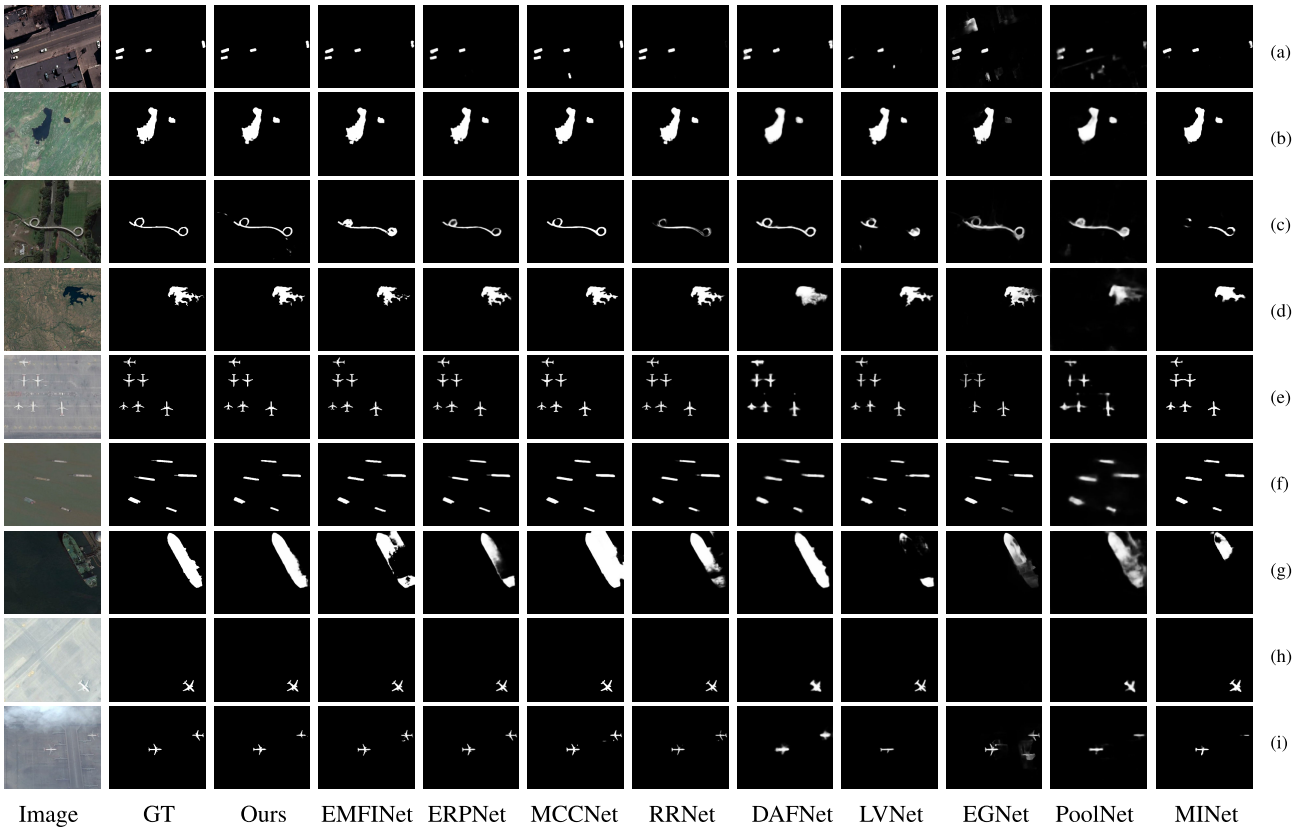


Fig. 7. Visualization comparison of ten SOTA methods, i.e., the proposed CRNet (ours), EMFINet, ERPNet, MCCNet, RRNet, DAFNet, LVNet, EGNet, PoolNet, and MINet. Clearly, our method achieves the best visual results. (a)–(i) Represent each row respectively, indicating the comparison of the results of different methods under the same test image.

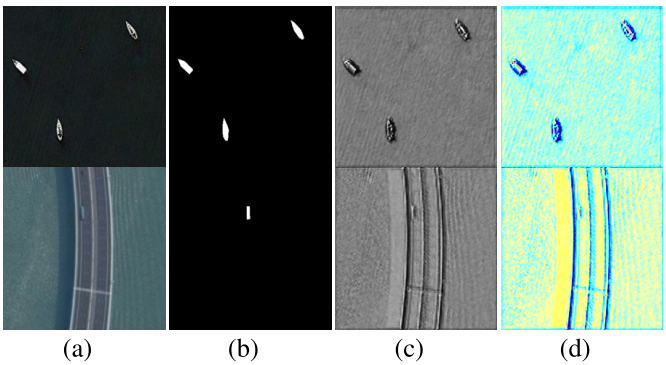


Fig. 8. Visualization results of ablation study. (a) ORSIs. (b) Ground truths. (c)–(d) Visualization and saliency maps produced by A^2.

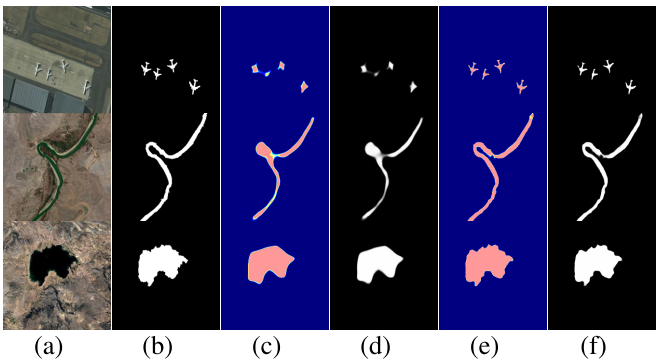


Fig. 9. Visualization results of ablation study. (a) ORSIs. (b) Ground truths. (c)–(d) Visualization and saliency maps produced by FF^4. (e)–(f) Visualization and saliency maps produced by FF^1.

c) Advantages in detecting multitarget scenarios: From (e), (f) rows of Fig. 7, it is clear that when several small targets are detected, the comparison algorithm either detects the background region inaccurately or misses all the relevant targets. Not only does the proposed CRNet detect every target but also it has sharp target edges, complete content, and excellent detection accuracy.

d) Advantages under imaging interference: In rows (g), (h), and (i) of Fig. 7, the background color is too close to the target color, and there is fog interference in ORSIs, which brings challenges to SOD. In such situations, ERPNet and MCCNet partially detect the salient targets while mistakenly emphasizing the harbor area. In contrast, the proposed CRNet outperforms other models. Specifically, CRNet can effectively suppress the surface and harbor areas, highlighting the ship targets. EGNet and LVNet cannot identify the disturbed salient points in scenes with fog interference, and other deep-learning-based comparison techniques also suffer from certain detection interference. Overall, the proposed CRNet can fully detect salient targets with accurate boundary details in the interference scenes.

The above quantitative and qualitative comparisons illustrate the effectiveness and superiority of the proposed CRNet model. They also significantly support the claim that our proposed CRNet model can provide precise and distinct SOD.

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TABLE VI
CAM ABLATION IN RFM ON THE EORSSD DATASET

Modules	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$Mae\downarrow$	$S_\alpha\uparrow$
full model	**0.8873**	0.9743	**0.0063**	**0.9370**
RF1 w/o CAM	0.8804	**0.9752**	0.0071	0.9333
RF1 w/o CAM + RF2 w/o CAM	0.8797	0.9745	0.0071	0.9327

TABLE VII
FURTHER VALIDATION OF CEM, RFM, AND LOSS ON THE EORSSD DATASET

Modules		$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$Mae\downarrow$	$S_\alpha\uparrow$
full model		**0.8873**	**0.9743**	**0.0063**	**0.9370**
CEM	w/o CEM(1)	0.8585	0.9636	0.0069	0.9359
	w/o CEM(2)	0.8670	0.9618	0.0064	0.9353
RFM	w/o RFM(3)	0.7737	0.9371	0.0079	0.9267
	w/o RFM(4)	0.8738	0.9695	0.0074	0.9277
	w/o RFM(5)	0.8860	0.9727	0.0065	0.9369
loss	w IoU	0.8739	0.9647	0.0075	0.9228
	w BCE	0.8803	0.9733	0.0070	0.9370

C. Ablation Study

In this section, we conduct ablation experiments to show the effectiveness of each module in CRNet. The significance of the proposed two modules in improving SOD has been experimentally verified. To determine: 1) the contribution of the CEM and RFM to the network model; 2) the efficiency of each component in the CEM and RFM; and 3) the efficiency of the hybrid loss function, we run the ablation experiments on the EORSSD dataset. For each experiment, we retrain the model with the same parameter settings and training data.

1) Contribution of CEM and RFM to the Network Model: We create two variations to evaluate the performance of CEM and RFM: 1) an encoder–decoder network (also known as "baseline") is a framework with only a Res2Net50 ($i = \{1, 2, 3, 4, 5\}$) decoder and 2) a baseline network with only CEM is a network framework with CEM to baseline for $i = \{1, 2\}$. Our model is "baseline + CEM + RFM" with RFM enhancing the network architecture for the features collected from $i = \{3, 4, 5\}$. In Table IV, we present the quantitative findings.

a) Effectiveness of CEM: Using attention mechanisms, CEM tries to capture channel interactions to emphasize relevant targets and improve feature representation. Table IV demonstrates that the baseline model with CEM performs better than the baseline model. $F_\beta^{\max}$ performance improves by 0.3%, $E_\xi^{\max}$ by 0.2%, S-measure by 4.1%, and Mae performance improves by 0.2%. In addition, Fig. 8 displays the feature mask results. Fig. 8 demonstrates that CEM successfully reduces background noise and concentrates attention on salient objects. It has been demonstrated that the proposed CEM works.

In the CEM module, the number of cross-channels and the residual structure are considered to be crucial. The effectiveness of the residual structure has been demonstrated in the community. Therefore, we primarily use an ablation comparison to investigate the impact of the chosen number of cross-channels. This submodule is used to capture channel interactions and enhance the representation capability of salient objects in the channel dimension. Table V presents a comparison between the adaptive method of selecting the number of cross-channels and a fixed method with three channels. The results reveal that the adaptive approach outperforms the fixed one by 2.1%, 0.9%, 17.5%, and 1.2% in $F_\beta^{\max}$, $E_\xi^{\max}$, Mae, and S_α, respectively. Clearly, this indicates that adaptive cross-channels can improve the feature extraction effectiveness of the CEM module. Consequently, we use an adaptive method to select the number of cross-channels for our study.

b) Effectiveness of RFM: RFM is developed to efficiently aggregate deep-level feature semantics. Table IV demonstrates that the performance of the baseline model using RFM and CEM is substantially improved compared with that of the baseline model with only CEM. The improvement in $F_\beta^{\max}$, $E_\xi^{\max}$, S_α, and Mae is 5.5%, 3.0%, 52.4%, and 1.6%, respectively. RFM collects contextual data to enhance the integrity of the salient targets in ORSIs, as can be observed in Fig. 9.

In the RFM module, we consider the nested CAM module to be a vital component. As shown in Fig. 4, the RFM module consists of two similar structures connected through resize. We define the structure before resize as RF1 and the structure after resize as RF2. To investigate the impact of nested CAM modules on the RFM module, we conducted split ablation experiments on the RFM module. The ablation experiments were set to "RF1 w/o CAM" and "RF1 w/o CAM + RF2 w/o CAM," representing only the removal of the CAM module in the RF1 structure and the removal of the CAM modules in both RF1 and RF2 structures. Table VI presents the experimental results. The comparison shows that the "full model" outperforms "RF1 w/o CAM + RF2 w/o CAM" by 0.9%, 12.7%, and 0.5% in $F_\beta^{\max}$, Mae, and S_α, respectively. By comparing the results of only removing the CAM module in RF1, completely removing the CAM module, and the complete model, we find that using CAM as the main structure of the RFM module is crucial for capturing useful features during feature remodeling and significantly improving the detection performance.

To verify the effectiveness of the two modules of CEM and RFM, we adopt a method for comparing CA and self-attention (SA). Specifically, we conducted the ablation experiments of "baseline + CA" and "baseline + SA," which were added in Table IV. The results showed that using the CA module as a shallow feature extraction model resulted in significantly lower accuracy than using the CEM module, proving that CEM has better feature representation capabilities from the channel perspective. On the other hand, using the SA module as a deep feature extraction model resulted in comparable accuracy to that of the RFM module. To further demonstrate the advantages of combining the CEM and RFM modules, we added four combined modes in Table IV: "baseline + CA + SA," "baseline + CA + RFM," "baseline + CEM + SA," and "baseline + CEM + RFM." In addition to the fact that "baseline + CA + RFM" surpassed the combination of "baseline + CEM + RFM" in $E_\xi^{\max}$ evaluation, the CEM and RFM modules generally showed significant advantages in terms of model accuracy. Therefore, the effectiveness of the CEM and RFM modules was verified by comparing CA and

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SA, and the efficiency of the combination of CEM and RFM was also verified by various combinations.

2) Effectiveness of CEM and RFM Scheduling: Five versions with the appropriate ablation upgrades for f^1–f^5 of the Res2Net50 output are carried out to confirm the validity and reasonableness of the CEM and RFM in terms of model organization. Table VII displays the ablation outcomes, where the values in parenthesis correspond to the respective f levels. Without CEM, the detection is substantially less accurate when compared with the accuracy of the entire model since it does not concentrate on mechanisms to capture channel interactions across layers. Without RFM, there is no feature association between the high-level semantic contexts acquired by reshaping, which makes it much more difficult to identify salient object locations. The experimental findings of the aforementioned variants lead to the conclusion that each component is genuine and reasonably arranged in its own right.

3) Effectiveness of Hybrid Loss Function: We use two different methods to experimentally show that the combination of BCE and IoU loss functions performs better. Using only the BCE loss function to replace the hybrid loss of the initial model is indicated by "w BCE," and using only the IoU loss function to replace hybrid loss of the initial model is denoted by "w IoU." Their quantitative results are presented separately in Table VII. This hybrid loss function integrates pixel-level and map-level supervision and better protects the integrity of the salient objects [59]. As can be seen in Table III, the results of the "full model" have a large advantage over the results of "w IoU" and "w BCE," thus demonstrating the effectiveness of the hybrid loss function.

D. Discussion

1) Failure Cases: Some failure cases are shown in Fig. 10. For some extremely challenging examples, our method still has biases and needs to be improved and optimized.

a) It is incomplete in the face of complex shadow problems: For example, in the first row of the figure, the shadow of the front half of the aircraft can suppress the discrimination, but the shadow of the second half of the aircraft is misidentified.

b) It is ambiguous in the face of low-contrast object boundaries: For example, in the second row of the figure, the boundary of the island is not clear, and the contrast between the edge content and the salient target is not high, and it is mistakenly identified as the main part.

c) It cannot determine the primary–secondary relationship when faced with multiple target problems: For example, in the third row in the figure, there are two more salient targets, and the more salient target cannot be confirmed in the salient comparison, resulting in detection redundancy.

2) Future Work: First, for the challenging examples mentioned in the failure cases, in addition to the fact that our model itself needs further optimization, we can try to use super-resolution to improve the resolution of the dataset to help strengthen shadows and edges and salient objects. The contrast of the subject can be optimized. Second, for multiple target problems, construct a specific dataset to strengthen the training

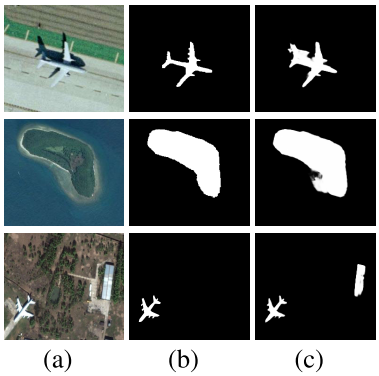


Fig. 10. Visual examples of failure cases. (a) Optical RSIs (ORSIs). (b) Ground truths. (c) Saliency maps produced by the proposed CRNet.

effect. At the same time, try to mix other spectral information (such as near-infrared channels invisible to humans) to obtain more comprehensive and richer information to help improve detection accuracy. In future work, we hope to build a model that takes into account both the detection effect and computational efficiency, so as to further realize the implementation of experiments.

V. Conclusion

This study builds an end-to-end CRNet for RSI-SOD. Specifically, two enhancement modules are combined in the backbone encoder to effectively suppress redundant background information. The CEM is used for high-resolution shallow features to direct local cross-channel interactions without dimensionality reduction in the production of CA maps. RFM uses feature transformation into a 2-D relationship graph for feature semantic reshaping and aggregation of global relationships. A multilevel cascade fusion approach is used in the decoder and combined with depth supervision to guide the final generation of salient graphs to further solve the scale change problem. Comparing our test performance to SOTA techniques, we also get better results. In the future, we intend to integrate CNN and transformer models to improve local features using chunked processing to increase network robustness and detection accuracy while preserving the structural integrity of the image.

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